Environmental Risk Assessment

Ocean temperature and pH dashboard, 2015 to 2023

1. Project overview

This project is an interactive Power BI report that assesses environmental risk from sea surface temperature (SST) and pH. It shows how that risk varied across seven marine locations from 2015 to 2023. The report combines ocean temperature, pH, marine heatwave occurrence, bleaching severity and species observations into one risk score, then shows where and when risk was highest.

The project demonstrates data cleaning in Power Query, a star-style data model, DAX measures and interactive dashboard design.

2. Research questions and short answers

- **Which locations have the highest environmental risk?** Caribbean Sea (average risk 11.9) and South China Sea (11.7). Red Sea has the lowest (10.4).
- **How do SST and pH vary across regions?** Average SST differs by less than 0.5 °C between locations (28.4 to 28.9 °C). Average pH is almost identical (8.04 to 8.06). Variation is mostly within locations, not between them.
- **What is the relationship between heatwaves and temperature?** In this dataset, a marine heatwave is recorded exactly when SST is 30 °C or above. Heatwave readings average 30.9 °C, compared with 28.1 °C otherwise.
- **Which locations show greater ecosystem stress?** Caribbean Sea and Hawaiian Islands have the highest share of High or Extreme readings (34% and 31%). South China Sea has the most heatwave readings (22%).
- **How can several indicators be combined?** By ranking SST and pH within the dataset and adding the two ranks into a single risk score (section 5).

3. Data source and coverage

- **Source:** Kaggle, "Shifting Seas: Ocean Climate and Marine Life" dataset, https://www.kaggle.com/datasets/atharvasoundankar/shifting-seas-ocean-climate-and-marine-life-dataset
- **Size:** 500 readings across 7 locations: Great Barrier Reef, Red Sea, South China Sea, Hawaiian Islands, Caribbean Sea, Galapagos and Maldives.
- **Period:** 1 January 2015 to 31 December 2023, with 55 or 56 readings in each year.
- **Hemisphere:** assigned from latitude. Northern for latitude 0 or above, Southern otherwise.

4. Attributes

Attribute	Type	Description	Example
Date	Date	Observation date (day/month/year in the source file)	31/12/2023
Location	Categorical	Reef location where the reading was taken	Maldives
Latitude	Decimal	Latitude of the observation point	20.0407
Longitude	Decimal	Longitude of the observation point	38.4861
SST (°C)	Decimal	Sea surface temperature in degrees Celsius	28.08
pH Level	Decimal	Ocean acidity measure	8.098

Attribute	Type	Description	Example
Bleaching Severity	Categorical	Bleaching intensity: Low, Medium or High. Blank in the source, shown as None	Low
Species Observed	Integer	Number of species observed in the reading	129
Marine Heatwave	Boolean	True when SST is 30 °C or above	True

Derived fields added during modelling: SST_C (renamed SST), pH type, pH Order, SST_Rating, pH_Rating, Risk Analysis (the risk score), Risk Rating, Risk Rank and Hemisphere.

5. Data preparation

- Removed a pivot table that had been pasted beside the data.
- Fixed the SST header encoding, renaming *SST (Â°C)* to *SST_C*.
- Read dates as day/month/year (English UK). About 40% of days are 12 or less, so the locale decides whether dates are read correctly.
- Labelled the 150 blank bleaching-severity cells as *None*. This is 30% of readings and is the main data gap.
- Replaced a constant "Alkaline" pH column with pH bands, since all readings fell in the same category.
- **Risk score.** Each reading's SST and pH are ranked by percentile within the dataset, from 1 to 10. Higher SST gives a higher temperature score. Lower pH gives a higher acidity score. The risk score is the sum, from 2 to 20. Ratings are Low (2 to 10), Moderate (11 to 14), High (15 to 18) and Extreme (19 to 20).
- **pH bands.** Normal (8.10 and above), Reduced (8.00 to 8.10), Acidifying (7.95 to 8.00) and Strongly acidifying (below 7.95). These cut-offs are analyst-defined, relative to the roughly 8.2 pre-industrial ocean pH.

6. Exploratory data analysis

6.1 Data quality

The file has 500 rows and no duplicate rows. Dates all parse, and each location keeps tight coordinates (within about 0.3 degrees). The only missing values are the 150 blank bleaching-severity entries. No values were removed.

6.2 Distributions

SST, pH and species are close to symmetric (skew near zero). Each has only two or three values flagged by the 1.5 × IQR rule. These are plausible readings and were kept.

Measure	Mean	Median	SD	Min	Max	Skew	IQR outliers
SST (°C)	28.54	28.52	1.43	23.64	33.21	0.08	3
pH level	8.050	8.052	0.056	7.872	8.195	-0.01	2
Species observed	120.5	120	20.5	54	171	-0.06	2
Risk score (2 to 20)	11.0	11	5.0	2	20	0.03	0

6.3 Categories

Bleaching severity: 150 blank (shown as None, 30%), Low 132, Medium 130, High 88. Marine heatwave: 73 readings (14.6%) are True. Risk rating: Low 233 (47%), Moderate 127 (25%), High 102 (20%), Extreme 38 (8%).

6.4 Relationships

Pair	Correlation (r)	Reading
SST and pH	-0.52	Warmer readings are more acidic

Pair	Correlation (r)	Reading
SST and species observed	-0.68	Warmer readings have fewer species
pH and species observed	0.33	More acidic readings have fewer species
SST and risk score	0.85	By construction: the score uses SST
pH and risk score	-0.85	By construction: the score uses pH

The risk score is built from SST and pH, so its correlations with them are expected and say nothing new. The useful relationships are between the environmental measures and species observed. Temperature and species show the clearest link (r = -0.68).

6.5 Seasonality

Average SST is highest in January (28.97 °C) and lowest in July (28.15 °C). Heatwave readings are most common in December (26%) and January (21%). With about 40 readings per calendar month across nine years, treat monthly patterns as indicative only.

7. Descriptive analysis: what happened

7.1 By location (sorted by average risk)

Location	Readings	Avg SST (°C)	Avg pH	Avg species	Avg risk score	High+Extreme %	Heatwave %
Caribbean Sea	70	28.7	8.040	118.5	11.9	34.3%	18.6%
South China Sea	73	28.9	8.046	116.9	11.7	30.1%	21.9%
Galapagos	65	28.4	8.047	120.4	10.9	23.1%	12.3%
Hawaiian Islands	71	28.4	8.049	122.3	10.9	31.0%	14.1%
Great Barrier Reef	87	28.5	8.053	122.4	10.6	25.3%	9.2%
Maldives	51	28.5	8.059	119.5	10.6	23.5%	13.7%
Red Sea	83	28.4	8.057	122.3	10.4	27.7%	13.3%

- Caribbean Sea has the highest average risk (11.9) and the highest share of High or Extreme readings (34%).
- Red Sea has the lowest average risk (10.4). Great Barrier Reef and Maldives follow at 10.6.
- The gap between the highest and lowest location is 1.5 points on a 2 to 20 scale. Differences are real but modest.

7.2 By year

Year	Readings	Avg SST (°C)	Avg pH	Avg species	Avg risk score	High+Extreme %	Heatwave %
2015	56	28.48	8.046	117.9	11.16	32.1%	17.9%
2016	56	28.17	8.075	121.7	9.09	14.3%	8.9%
2017	55	28.76	8.036	120.6	11.85	32.7%	14.5%
2018	55	28.67	8.059	123.4	10.96	27.3%	18.2%
2019	56	28.55	8.046	122.5	11.32	28.6%	10.7%
2020	55	28.87	8.041	117.0	12.33	40.0%	23.6%
2021	56	28.47	8.054	122.5	10.59	26.8%	12.5%
2022	55	28.65	8.045	116.1	11.36	27.3%	14.5%
2023	56	28.23	8.047	122.5	10.30	23.2%	10.7%

- 2020 was the highest-risk year: average risk 12.3, and 40% of readings High or Extreme.
- 2016 was the lowest: average risk 9.1, with 14% High or Extreme.
- Average SST shows no trend over the nine years (0.0002 °C per year). Average risk rises by 0.04 points per year, which is flat once the noise is considered.

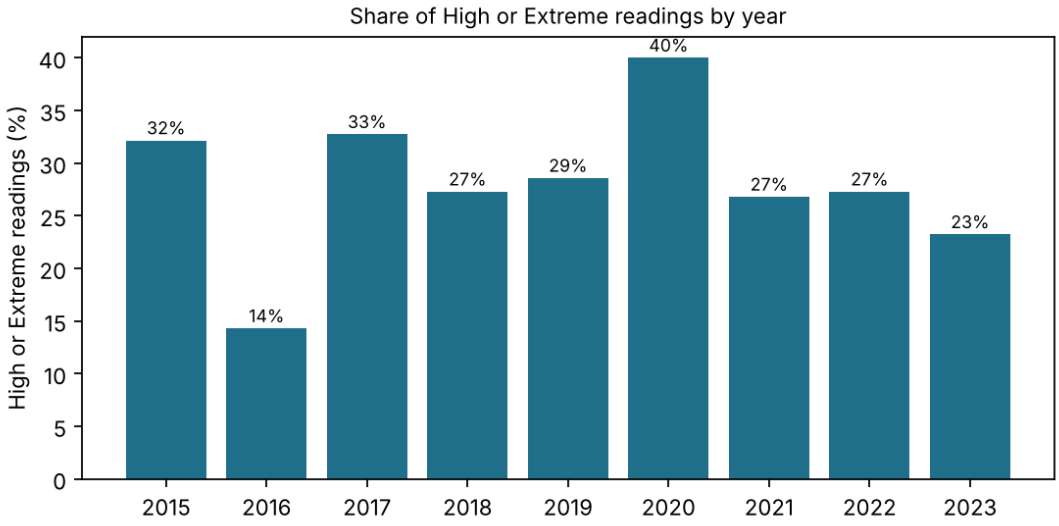


Figure 1. Share of readings rated High or Extreme, by year.

8. Diagnostic analysis: why risk varies

8.1 Marine heatwaves

Group	Readings	Avg SST (°C)	Avg pH	Avg species	Avg risk score
No marine heatwave	427	28.14	8.06	124.4	9.8
Marine heatwave	73	30.85	8.00	97.6	17.8

Heatwave readings are about 2.7 °C warmer, have 27 fewer species on average, and sit at a risk score nearly twice as high (17.8 against 9.8). Because a heatwave is defined by SST at 30 °C or above, this result is best read as part of the temperature finding.

8.2 Temperature and species observed

Species observed falls as SST rises across the dataset (r = -0.68). The same direction holds inside every location, with correlations between -0.62 and -0.73. On average, each extra degree Celsius goes with about 10 fewer species. Temperature and pH are also negatively related (r = -0.52, and between -0.41 and -0.64 within locations).

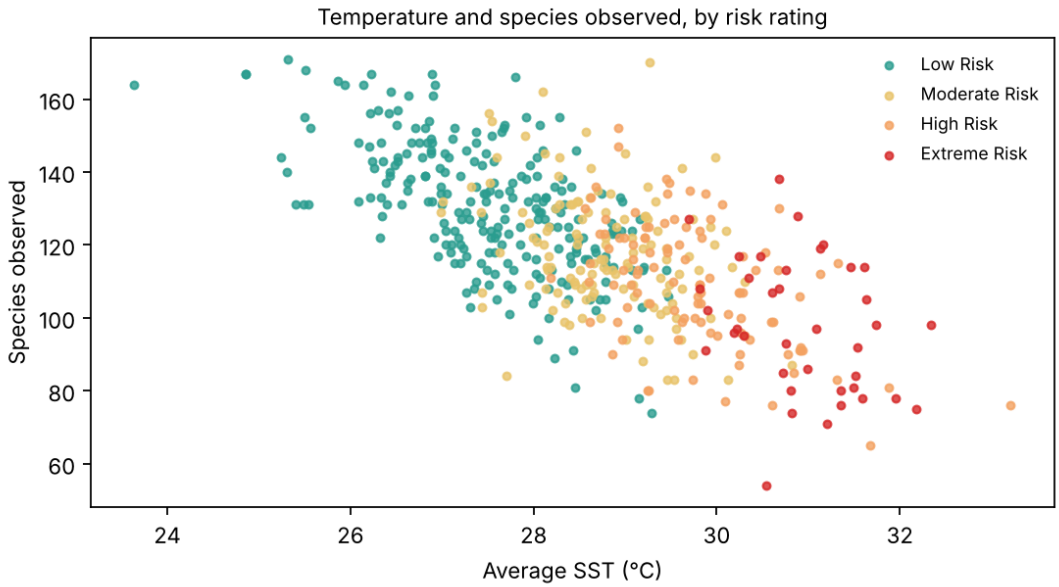


Figure 2. Each point is one reading. Colour shows its risk rating.

8.3 Risk ratings and species observed

Risk rating	Score range	Readings	Avg SST (°C)	Avg pH	Avg species
Low Risk	2 to 10	233	27.48	8.091	130.5
Moderate Risk	11 to 14	127	28.77	8.037	118.0
High Risk	15 to 18	102	29.76	8.005	109.2
Extreme Risk	19 to 20	38	30.94	7.957	97.6

Average species falls at each step up in rating, from 130 at Low to 98 at Extreme. This supports the risk score as a useful summary, though the score is partly built from the same temperature and pH values.

8.4 Bleaching severity

Bleaching severity	Readings	Avg SST (°C)	Avg pH	Avg species
None	150	28.44	8.051	123.7
Low	132	28.52	8.050	119.0
Medium	130	28.61	8.050	119.5
High	88	28.62	8.047	118.7

Average SST barely changes across bleaching levels (28.4 to 28.6 °C), and species observed are slightly higher where no bleaching is recorded (123.7, against 118.7 to 119.5 for the bleaching levels). This dataset does not show that warmer reefs bleach more. With 30% of bleaching records missing, this result is a starting point for further testing.

8.5 Bleaching as an outcome check

Bleaching is an outcome, so it is not part of the risk score. Instead, the score is tested against it. The 150 blank bleaching cells are unrecorded, not confirmed as no bleaching, so every rate below uses the 350 readings with a recorded severity.

Risk rating	Recorded readings	High bleaching	High share
Low	158	37	23.4%
Moderate	89	22	24.7%
High	74	21	28.4%
Extreme	29	8	27.6%

The High bleaching share stays between 23% and 28% across risk ratings. A chi-square test across all three severity levels finds no link ($p = 0.88$), and the rank correlation between severity and risk score is $r = 0.03$ ($p = 0.63$). On marine heatwave days the High share is 26.3% (57 readings), against 24.9% on other days (293 readings). That gap is too small to read. In this dataset, bleaching does not change with heat. Complete bleaching records would allow a firmer test.

9. Predictive analysis

A linear regression predicts species observed from SST. The model was fitted on 2015 to 2021 (389 readings) and tested on 2022 to 2023 (111 readings, held out). Its equation is roughly $species = 399.8 - 9.8 \times SST$, with an R-squared of 0.46 on the training data.

- Test error (RMSE): 15.0 species, compared with 20.4 for predicting the average. That is about 26% lower.
- This is a single-predictor model. It demonstrates the method and is not a forecast.
- No causal claim is made. Other factors, such as local pollution, fishing and depth, are not in the data.

10. Prescriptive analysis: recommendations

- **Focus monitoring on Caribbean Sea and South China Sea.** They have the highest average risk and the most High or Extreme readings.
- **Set a heat alert at 30 °C.** That is the threshold the data uses for heatwaves, and heatwave readings carry an average risk score of 17.8, against 9.8 otherwise.
- **Watch Hawaiian Islands and Maldives.** Both rose between 2020 and 2021 (11.2 to 13.5 and 10.0 to 11.6), while the other five locations fell.
- **Complete the bleaching records.** With 30% of severity values missing, a heat-bleaching link is not yet testable. Consistent recording is the first priority.
- **Measure anomalies, not just absolute temperature.** Reefs tolerate different temperatures, so each reading should also be compared with its own location's baseline.
- **Track species alongside temperature.** In this data, species fall by about 10 per °C.

11. Data model and DAX

- **Fact table:** Bleaching_events, one row per reading, with the calculated score columns.
- **Date table:** calendar, marked as the date table and linked to the fact table on Date (many to one, single direction).
- **Measures table:** DAX_table, holding every measure so they sit apart from the data.

Key measures

Readings; Average SST; Average Ph; Avg Species; Average Risk rating; Heatwave %; High+Extreme Readings; High+Extreme %; High+Extreme % PY; High+Extreme Change (pts); Risk PY; Risk Change; Risk Change %; Species PY; Species Change %.

Year-over-year measures use SAMEPERIODLASTYEAR on the calendar table. They work when one year is selected in the Year slicer.

12. Dashboard pages

- **Overview:** headline KPI cards, a bubble map of reef risk, a risk-rating mix by year, and the average risk trend.
- **Drivers:** SST against species scatter, species by risk rating, species by pH band, and average risk by location.
- **2020 vs 2021:** year-over-year cards, a location matrix, a monthly trend and High+Extreme % by location.
- **Location Deep Dive:** a drill-through page. SST over time against the 30 °C threshold, the High and Extreme readings, and the bleaching mix.
- **Tooltips:** definitions of High+Extreme % and the risk score.

13. Scope and next steps

- **Scope of the score.** Ratings compare readings within this dataset, so they rank risk rather than set a coral-safety threshold. Next step: compare against published ecological limits.
- **Heatwave flag.** A heatwave is defined as SST of 30 °C or above, so it tracks the same signal as temperature. Next step: define heatwaves against a baseline for each location.
- **Bleaching records.** Severity is recorded for 350 of 500 readings, and bleaching is reported alongside the score rather than inside it (section 8.5). Next step: complete bleaching records, which would allow a formal test of heat and bleaching.
- **Sample size.** About 55 readings per year reveal clear patterns. Year-to-year changes are best read as direction rather than proof. Next step: more readings per location and year.
- **Dataset.** The patterns are unusually regular, so the project is best read as a method demonstration. Next step: rerun the same workflow on observed monitoring data.

14. Project outcome

Risk was highest in Caribbean Sea and South China Sea, and in 2020. Risk fell in 2021 at five of the seven locations, but rose at Hawaiian Islands and Maldives, and rebounded in 2022. Across all readings, species observed fall as SST rises and as pH becomes more acidic. These are patterns in one dataset. Monitoring should focus on heatwave-prone reefs, and more complete bleaching records would allow a firmer test of heat-driven bleaching.